Department of Computer Engineering

Technical Interview Question & Model Answer Bank

Academic Year	**2026-2027**	**Class**	**4th year**
Subject/Expert Area	**DBMS & SQL**		
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Q. No.	Interview Question	Answer	Example
1	What is a Database?	A database is an organized collection of related data that can be stored, accessed, and managed efficiently.	A college database may store student, course, faculty, and examination information.
2	What is DBMS?	A DBMS (Database Management System) is software used to create, store, retrieve, update, and manage data in a database.	MySQL, Oracle, PostgreSQL, and SQL Server are DBMSs.
3	What is RDBMS?	RDBMS stores data in the form of related tables consisting of rows and columns and generally uses SQL for data manipulation.	MySQL, Oracle Database, PostgreSQL, and SQL Server are RDBMSs.
4	What is the difference between DBMS and RDBMS?	DBMS is a general database management system, while RDBMS organizes data into related tables and supports relational concepts such as keys and relationships.	A relational database can have Student and Department tables connected through Dept_ID.
5	What are the advantages of using a DBMS?	Major advantages include reduced redundancy, improved data integrity, security, concurrency control, backup/recovery, and easier data sharing.	Multiple departments can access the same student database without maintaining separate copies.
6	What is data redundancy? How can it be reduced?	Data redundancy means unnecessary duplication of the same data. It can be reduced through proper database design and normalization.	Storing a student's department name repeatedly in every student record creates redundancy.
7	What is data integrity?	Data integrity means maintaining	A foreign key can prevent a

		the accuracy, consistency, and validity of data in a database.	student from being assigned to a department that does not exist.
8	What is a database schema?	A schema is the logical structure or design of a database, including tables, attributes, relationships, and constraints.	Student(RollNo, Name, DeptID) represents part of a database schema.
9	What is a database instance?	An instance is the actual data stored in the database at a particular point in time.	The current rows stored in the Student table constitute its current instance.
10	What is a table in a relational database?	A table is a collection of related data organized into rows and columns.	Student(RollNo, Name, Branch) is a table with three columns.
11	What is a tuple?	A tuple is a single row or record in a relational table.	(101, 'Rahul', 'CSE') is one tuple of the Student table.
12	What is an attribute?	An attribute is a column or property that describes an entity or relation.	Name, RollNo, and Branch are attributes of Student.
13	What is an entity?	An entity is a distinguishable real-world object about which data is stored.	A student, employee, or department can be an entity.
14	What is a relationship in an ER model?	A relationship represents an association between two or more entities.	A Student enrolls in a Course.
15	What is a primary key?	A primary key is an attribute or combination of attributes that uniquely identifies each row in a table. It cannot contain NULL values.	RollNo can be the primary key of the Student table.
16	What is a candidate key?	A candidate key is a minimal attribute set that can uniquely identify a record. One candidate key is selected as the primary key.	Both RollNo and Email may uniquely identify a student; both can be candidate keys.
17	What is a super key?	A super key is any set of one or more attributes that uniquely identifies a tuple. It may contain unnecessary attributes.	{RollNo} and {RollNo, Name} can both be super keys.
18	What is a foreign key?	A foreign key is an attribute that refers to a primary key or candidate key in another table and is used to establish a relationship.	Student.DeptID can reference Department.DeptID.
19	What is a composite	A composite key is a key	(StudentID, CourseID) can

	key?	consisting of two or more attributes used together to uniquely identify a record.	uniquely identify a student's course enrollment.
20	What are integrity constraints?	Integrity constraints are rules that maintain the correctness and consistency of database data.	PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, and CHECK are examples.
21	What is normalization?	Normalization is the process of organizing data to reduce redundancy and prevent insertion, deletion, and update anomalies.	Separating Student and Department information into different tables is a normalization approach.
22	What is 1NF?	A relation is in First Normal Form when each attribute contains atomic values and there are no repeating groups.	Instead of storing Phone = 9876, 8765 in one cell, store phone numbers separately.
23	What is 2NF?	A relation is in 2NF when it is in 1NF and every non-key attribute is fully dependent on the entire candidate key.	In an enrollment table with (StudentID, CourseID), CourseName should not depend only on CourseID.
24	What is 3NF?	A relation is in 3NF when it is in 2NF and has no transitive dependency of non-key attributes on the key.	If StudentID → DeptID and DeptID → DeptName, separating Department information helps achieve 3NF.
25	What is BCNF?	BCNF is a stronger form of 3NF where every determinant must be a candidate key.	If X → Y, then X must be a candidate key for the relation to satisfy BCNF.
26	What are insertion, deletion, and update anomalies?	These are problems caused by poor database design and redundancy. They can cause inconsistent or lost information.	Updating a department name in several student records may lead to inconsistent department names.
27	What is SQL?	SQL (Structured Query Language) is used to define, manipulate, retrieve, and control data in relational databases.	SELECT * FROM Student; retrieves student records.
28	What are the major categories of SQL commands?	SQL commands are commonly classified as DDL, DML, DQL, DCL, and TCL.	CREATE → DDL, INSERT → DML, SELECT → DQL, GRANT → DCL, COMMIT → TCL.
29	What is DDL?	DDL (Data Definition Language) is used to define or modify database structures.	CREATE TABLE Student (...);

30	What is DML?	DML (Data Manipulation Language) is used to insert, update, and delete data.	INSERT INTO Student VALUES (101,'Rahul','CSE');
31	What is DQL?	DQL (Data Query Language) is used to retrieve data from a database. SELECT is commonly classified as DQL.	SELECT Name FROM Student;
32	What is the difference between DELETE, TRUNCATE, and DROP?	DELETE removes selected rows, TRUNCATE removes all rows while retaining the table structure, and DROP removes the table itself.	DELETE FROM Student WHERE RollNo=101; removes one row, while DROP TABLE Student; removes the table.
33	What is the SELECT statement?	SELECT is used to retrieve data from one or more tables.	SELECT Name, Branch FROM Student;
34	What is the WHERE clause?	WHERE filters individual rows according to a specified condition.	SELECT * FROM Student WHERE Branch='CSE';
35	What is the difference between WHERE and HAVING?	WHERE filters rows before grouping, while HAVING filters groups after GROUP BY.	WHERE Salary > 30000 filters employees; HAVING AVG(Salary) > 50000 filters departments.
36	What is GROUP BY?	GROUP BY groups rows having the same values so aggregate functions can be applied to each group.	SELECT DeptID, COUNT(*) FROM Employee GROUP BY DeptID;
37	What is ORDER BY?	ORDER BY sorts query results in ascending or descending order.	SELECT * FROM Employee ORDER BY Salary DESC;
38	What are aggregate functions in SQL?	Aggregate functions perform calculations on a set of rows and return a single result for each group.	COUNT(), SUM(), AVG(), MIN(), and MAX().
39	What is a JOIN?	A JOIN combines rows from two or more tables based on a related column or condition.	Joining Student and Department using DeptID.
40	What is the difference between INNER JOIN and LEFT JOIN?	INNER JOIN returns only matching rows from both tables. LEFT JOIN returns all rows from the left table and matching rows from the right table.	A LEFT JOIN can display all students, including students who have no matching department record.
41	What is a subquery?	A subquery is a query written inside another SQL query.	SELECT Name FROM Employee WHERE Salary > (SELECT AVG(Salary) FROM Employee);
42	How would you find	Use a subquery to calculate the	SELECT * FROM Employee

	employees earning more than the average salary?	average salary and compare each employee's salary with it.	WHERE Salary > (SELECT AVG(Salary) FROM Employee);
43	What is a view?	A view is a virtual table based on the result of a SQL query. It generally does not store the underlying data separately.	CREATE VIEW CSE_Students AS SELECT * FROM Student WHERE Branch='CSE';
44	What is an index?	An index is a database structure that improves the speed of data retrieval from a table, usually at the cost of additional storage and maintenance.	Creating an index on Employee(EmpID) can speed up searches using EmpID.
45	What are the advantages and disadvantages of indexes?	Indexes can make SELECT queries faster, but they require storage and can make INSERT, UPDATE, and DELETE operations slower because indexes must also be maintained.	An index on Email helps quickly find a student by email but must be updated when email values change.
46	What is a transaction?	A transaction is a logical unit of database operations that should be completed successfully as a whole or rolled back.	Transferring ₹1,000 from Account A to Account B involves debit and credit operations in one transaction.
47	What are ACID properties?	ACID properties ensure reliable transaction processing, guarantee data reliability and integrity in databases. ACID stands for Atomicity, Consistency, Isolation and Durability • Atomicity: All operations must succeed or none are applied. • Consistency: Ensures the database remains in a valid state. • Isolation: Prevents interference from other concurrent transactions. **Durability: Guarantees the results are permanently stored.**	A bank transfer should either complete both debit and credit operations or complete neither.
48	What is atomicity?	Atomicity means a transaction is treated as an indivisible unit: either all its operations are completed or none are.	If money is debited but the credit operation fails, the transaction should be rolled back.
49	What is the difference between COMMIT	COMMIT permanently saves transaction changes, while	UPDATE Employee SET Salary=50000 WHERE

	and ROLLBACK?	ROLLBACK undoes uncommitted changes.	EmpID=101; COMMIT; permanently saves the change.
50	How can you find the second-highest salary in SQL?	A subquery or ranking function can be used to identify the second-highest distinct salary.	SELECT MAX(Salary) FROM Employee WHERE Salary < (SELECT MAX(Salary) FROM Employee);
51	What is CRUD ?	CRUD stands for which are the four fundamental operations on the database - Create, Read, Update, Delete,: • Create: Use the INSERT INTO statement to add new records to a table. • Read: Use the SELECT statement to retrieve data from a table. • Update: Use the UPDATE statement to modify existing records. Delete: Use the DELETE FROM statement to remove records.	
52	What is schema in DBMS ?	In DBMS, data is organized as Tables, views and relationship. Schema defines the tables, columns, along with their constraints, keys and relationship	
53	What is a trigger?	A trigger is a procedural code that gets executed automatically in response to certain events on a particular table or view.	
54	What is denormalization?	Denormalization combines normalized tables to improve performance. It is used when read performance is critical, and the application can handle data redundancy and potential update anomalies.	
55	What is NULL in SQL ?	NULL represents missing value, it is not 0 for numeric or blank for strings	EMP_Name is NULL doesn't means that the employee dosent have a name , it means that the name value is not stored in the DB

Scenario Based Questions :

Sr. No.	Question	Model Answer
1	A university has a STUDENT table and an ENROLLMENT table. A student cannot be deleted from the STUDENT table because he/she has enrolled for some courses and the corresponding data is stored in enrollment table. Why is the DBMS preventing the deletion? What are the possible solutions?	Since there is a Foreign key constraint set between the two tables . This is to preserve the referential integrity,. Possible constraint are : ON DELETE CASCADE, ON DELETE SET NULL, etc.
2	Two HR employees open the same employee record at almost the same time. Both see the salary as ₹40,000. Employee A changes it to ₹45,000, while Employee B changes it to ₹42,000. Finally, the database contains ₹42,000. What problem has occurred? Which DBMS concept is used to handle such situations?	These are Concurrent transactions and issue is , lost update problem. Solution for concurrency control is use of locking – the record should be locked before the update.
3	A customer transfers ₹10,000 from Account A to Account B. The system successfully deducts ₹10,000 from Account A, but before adding it to Account B, the server crashes. How can we rectify this?	The transfer should be treated as one transaction. If the server crashes before completing the transfer, then the DBMS uses its transaction log/recovery to ROLLBACK the ₹10,000 deduction from Account A, restoring the database to its previous consistent state.
4	A university has 1 million student records stored in the STUDENT table. Whenever an administrator searches for a student using student_id, the query takes several seconds. How can we make the query perform better?	An index should be created on the student_id column.
5	A college wants to maintain a database for students, courses, and course registrations. • A student can register for multiple courses, and a course can have multiple students. • For each registration, the college needs to store the registration date and grade. Design the database using three tables and identify the primary . foreign keys and relations.	Student(Student_ID PK, Student_Name, Email) Course(Course_ID PK, Course_Name, Credits) Registration(Student_ID FK, Course_ID FK, Registration_Date, Grade) Student_ID + Course_ID → Composite Primary Key in Registration. Student ↔ Course is Many-to-Many (M:N), realized through the Registration table.